GenRecon End-to-End Pipeline

`run_full_pipeline.py` – stages, output layout and CLI reference

- **1. Overview** – what the pipeline does, its modes and inputs
- **2. Stages** – every stage, in execution order
- **3. Output folder structure** – `runs/<scene_name>/` for both run types
- **4. CLI arguments** – all flags, defaults and validation rules

1. Overview

`run_full_pipeline.py` turns a set of photos (or a video, or a rectified stereo pair) of a real scene into a physics-ready Isaac Sim scene and a robot demo video. VGGT-Omega predicts camera poses and depth; the objects are segmented and reconstructed in 3D, converted to USD assets, composed into a scene, checked for inter-object penetration, and finally used in a robot demo. It is a Python port of `run_full_pipeline.sh`: every stage except Isaac Sim runs in-process as a typed function call (`genrecon/pipeline/stages.py`); Isaac Sim steps run as subprocesses in their own `isaacsim_venv` environment.

Run modes

Mode (`--mode`)	Object source	What happens	Final deliverable
pick-and-place (default)	Stage 0b: Qwen3-VL discovers object labels + boxes (conditioned on `--classes` if given).	A pick/place pair is chosen (random, from `--pick_place_target`, by the scene agent, or from a `--mimic-robot` video caption). Fast path: only the objects are reconstructed (Stage P1–P4).	`pick_place/pick_place.mp4`
full-scene	Grounding DINO with the required `--classes`.	Full GenRecon reconstruction of the whole scene (background, floor, objects) and composed USD scene.	`stage_3_genrecon/shapes/glb/scene.usda`, `final_objects/`
full-scene-with-robot	As full-scene.	Full-scene, then a robot-collision demo against `--robot-target` (random class if omitted).	As full-scene + `robot_collision/robot_collide.mp4`

Inputs

- **Image folder** (default): `run_full_pipeline.py <image_folder> <scene_name>`.
- **Video** (`--video`): frames are sampled with ffmpeg at 5 fps, resized to 960x540 into a temporary folder (deleted afterwards) and fed to Stage 0.
- **Stereo pair** (`--stage-0-use-stereo`): FoundationStereo replaces VGGT-Omega in Stage 0 (pick-and-place only; needs one left and one right image plus a ZED .conf calibration).
- **Robot episode video** (`--mimic-robot`): a video of a real pick-and-place; Qwen3-VL captions it and the caption selects the picked object and its target.

Pipeline flow

```
Common      Stage 0 -> [0b discovery] -> [0c caption] -> Stage 1 segmentation -> Stage 2 MV-SAM3D
                                                          (Stage 1-2 run only if there are classes)

pick-and-place (fast path, returns after P4)
            [P0 scene agent] -> P1 align meshes -> P2 convert assets -> P3 compose scene
                             -> P3b penetration check/fix -> P4 Franka pick-and-place video

full-scene / full-scene-with-robot
            3 reconstruct -> 4 reprojection check -> 5 object meshes -> 6 floor + friction
            -> 7 mesh->GLB -> 8 convert assets -> 9 compose scene -> 9b penetration check
            -> [10 GLB bake, disabled] -> 11 final objects -> [12 robot collision demo]
```

Runtime behaviour

- **Resuming:** `--start-from-stage N` skips earlier stages and reuses their outputs on disk (Stage 0b/0c reuse `discovered_*.json` / `caption.json` so the class list cannot drift); `--stop-after-stage N` exits cleanly after stage N.
- **Logs:** `pipeline.log` (everything, including mirrored per-stage logs) and `debug_config.log` (Isaac Sim subprocess configuration) in the run folder.
- **Failure:** any exception is logged with a traceback and the process exits with status 1.

2. Stages

Stage names match the `with stage(...)` blocks in `main()` and the logs. “Isaac” marks steps that run as a subprocess in `isaacsim_venv`.

2.1 Common stages (all modes)

Stage	What it does	Reads	Writes
0 VGGT-Omega export	Loads `vggt_omega_1b_512.pt`, predicts poses + depth at 512 px, gravity-aligns the scene (`--rotate-horizontal-deg` adds a yaw), exports a COLMAP dataset and per-frame depth/confidence (point cloud capped at 1M points, filtered by `--vggt-conf-thres`). With `--stage-0-use-stereo`, FoundationStereo is used instead.	image folder / video frames / stereo pair	`stage_0_vggt/{rgb,depth,colmap}`
0b Object discovery (pick-and-place)	Qwen3-VL-8B-Instruct returns object labels and boxes (left/right view) for the table scene; with `--classes` it looks for exactly those objects. Result is cached.	`stage_0_vggt/rgb`	`discovered_classes.json`, `discovered_boxes.json`
0c Episode captioning (`--mimic-robot`)	Samples `--caption-num-frames` frames from the episode video; Qwen3-VL names the picked object, the hand used and the placement location. The caption chooses the pick/place pair; if the picked object is not among the detected classes, a random pair is used.	episode video, detected classes	`caption.json`
1 Segmentation	Prompted SAM2 segmentation per class (boxes from Stage 0b in pick-and-place, Grounding DINO in full-scene), lifted to 3D with the Stage 0 depth (`--depth_conf_thres`, `--depth_edge_rtol`). Masks failing the hull-consistency check are rejected. Produces per-class masks and point clouds plus a background cloud. Runs only if classes exist.	`stage_0_vggt`, classes	`stage_1_segmentation/` (`labels.json`, per-class dirs, `background/`)
2 MV-SAM3D reconstruction	Multi-view SAM3D generates one textured mesh per class: shape (stage 1 steps) then texture (stage 2 steps), using the best `--mvsam3d-top-k-views` views, optional shape-latent alignment across views, fixed seed. Meshes are collected and previews written (aligned `*_MESH.ply` and scene-scaled `*_PCL.ply`).	`stage_0_vggt`, `stage_1_segmentation`	`stage_2_mv_sam3d/<label>/{mesh.glb,*_MESH.ply,*_PCL.ply}`

2.2 Pick-and-place path (Stage P*)

After Stage 2 the pipeline takes the fast path and ends after P4; Stages 3–12 are not run.

Stage	What it does	Writes
P0 Scene agent (`--ai-scene-agent`)	Interactive Ollama agent (`--scene-agent-ollama-model`) picks pick/place targets, approach side, clearance, gripper width, lighting and camera; passed on as extra args to P4.	`pick_place/scene_spec.json`
P1 Align meshes	Scales/aligns every class mesh to the real scene using its Stage 1 point cloud as scale reference; sanitised label directories.	`pick_place/glb/<label>/mesh.glb`
P2 Convert assets (Isaac)	`convert_asset.py` builds `asset.usd` per object with convexDecomposition collision.	`glb/<label>/asset.usd`, `convert_asset.log`
P3 Compose scene (Isaac)	`compose_isaac_scene.py` assembles the objects on a table prop (optional kitchen room via `--room`).	`glb/scene.usda`, `compose_isaac_scene.log`
P3b Penetration check (Isaac)	Detects inter-object penetration through PhysX contacts (`--penetration-check`). With `--penetration-fix` it writes +z lifts that clear pairs back into `scene.usda` (backup `scene.pre_penetration_fix.usda`); with `--penetration-settle` dynamic objects are then settled under gravity and the settled poses are written back.	`penetration.json`, `penetration.log`
P4 Franka pick-and-place (Isaac)	`demo_franka_pickplace.py`: Franka picks the target and places it on the place object (with clearance) or at `--place-offset`.	`pick_place.mp4`, `pick_place_scene.usda`, `pick_place.log`

2.3 Full-scene path (Stages 3–12)

Stage	What it does	Writes
3 Reconstruct scene	`reconstruct_scene.py`: GenRecon chunked scene reconstruction from the Stage 0 views (`--num_imgs_per_scene`, `--fix_num_chunks`, `--max_chunks_per_group`, `--max_inflated_voxels`), using the Stage 1 masks when classes exist.	`stage_3_genrecon/mesh.ply`, chunk files, `cameras.json`
4 Reprojection validation	Renders synthetic views of the reconstruction at the input cameras for visual comparison.	`stage_4_reproject_synth/{synth_views,compare_views}`
5 Object mesh extraction	Cascading per-object extraction from the scene mesh using each object's point cloud and masks; remaining geometry becomes the background; floating fragments near each object are removed.	`shapes/<label>_mesh.ply`, `_floaters.ply`, `background_mesh.ply`
6 Floor + friction	Segments the floor out of the background; infers per-object friction from a material table (`--friction-table-path`) using an Ollama model (`--ollama-model`).	`shapes/floor_mesh.ply`, `friction_assignments.json`
7 Mesh to GLB	`mesh_to_glb.py` converts meshes to GLB and substitutes each object with its aligned MV-SAM3D mesh.	`shapes/glb/<label>/mesh.glb`
8 Convert assets (Isaac)	As P2, plus friction table with max combine mode.	`shapes/glb/<label>/asset.usd`
9 Compose scene (Isaac)	Composes background, floor and objects (no table prop; optional `--room`).	`shapes/glb/scene.usda`
9b Penetration check (Isaac)	As P3b, on the full scene.	`stage_3_genrecon/penetration.json`
10 Chunked GLB bake	`chunked_to_glb.py` bakes a textured `scene.glb`. Currently disabled in code (`RUN_GLB = False`).	`scene.glb` (when enabled)
11 Final objects	Collects per-object deliverables: mesh, MV-SAM3D GLB and detection image.	`final_objects/<label>/`, `background_mesh.ply`
12 Robot collision (Isaac)	`demo_robot_collide.py`: the robot is driven into `--robot-target`. Needs Isaac (not with `--skip_isaac`).	`robot_collision/robot_collide.mp4`, `.usda`, `.log`

3. Output folder structure

Everything is written to `runs/<scene_name>/`. Labels containing spaces are kept raw in `stage_2_mv_sam3d/` and `final_objects/` but sanitised (space/slash to underscore) in `stage_1_segmentation/`, `pick_place/glb/` and `shapes/`.

3.1 Full-scene-with-robot (example: runs/kitchen_floor)

```
runs/kitchen_floor/
|-- pipeline.log                    full pipeline log
|-- debug_config.log                Isaac Sim subprocess config
|-- stage_0_vggt/
|   |-- rgb/                        input frames
|   |-- depth/                      frame_*_depth.npy, frame_*_depth_conf.npy
|   `-- colmap/                     cameras.txt, images.txt, points3D.{txt,ply}
|-- stage_1_segmentation/
|   |-- labels.json  hull_consistency_rejections.json  segmentation.log
|   |-- <label>/                    banana, bowl, plastic_cup, white_chair
|   |   |-- mask_bin/ mask_overlay/ mask_proj/    binary masks, overlays, projections
|   |   |-- mask_bin_rejected/ mask_overlay_rejected/   (only if some were rejected)
|   |   `-- point_cloud/<label>.ply
|   `-- background/point_cloud/
|-- stage_2_mv_sam3d/
|   |-- mv_sam3d.log
|   |-- input/                      images, per-class masks, da3_output.npz
|   `-- <raw label>/                mesh.glb, <label>_MESH.ply, <label>_PCL.ply
|-- stage_3_genrecon/
|   |-- mesh.ply  clean_points.ply  object_source_mesh.ply  cameras.json  args.json
|   |-- chunk_inputs.pt  to_glb_inputs.pt  chunk_transforms.json  coords_000..015.ply
|   |-- penetration.json  penetration.log  compose_isaac_scene.log  convert_asset.log
|   `-- shapes/
|       |-- <label>_mesh.ply  <label>_floaters.ply
|       |-- background_mesh.ply  floor_mesh.ply  friction_assignments.json
|       `-- glb/
|           |-- scene.usda  scene.pre_penetration_fix.usda
|           `-- <label>/ background/ floor/    asset.usd, manifest.json, mesh.glb, textures/
|-- stage_4_reproject_synth/
|   |-- synth_views/                re-rendered views
|   `-- compare_views/              input vs. synthetic comparisons
|-- final_objects/
|   |-- background_mesh.ply
|   `-- <raw label>/                mesh.ply, trellis_mesh.glb, detection.png
`-- robot_collision/
    `-- robot_collide.mp4  robot_collide_scene.usda  robot_collide.log
```

3.2 Pick-and-place (example: runs/kitchen_plates)

```
runs/kitchen_plates/
|-- pipeline.log  debug_config.log
|-- discovered_classes.json         Stage 0b labels (cached for --start-from-stage reruns)
|-- discovered_boxes.json           Stage 0b boxes per label (left/right view)
|-- caption.json                    only with --mimic-robot
|-- stage_0_vggt/                   rgb/  depth/  colmap/           (as above)
|-- stage_1_segmentation/           labels.json, <label>/ ...       (as above)
|   `-- white_bowl  white_plate_1  white_plate_2  silver_knife_1  silver_knife_2  glass_dish
|-- stage_2_mv_sam3d/               <raw label>/ ("white plate 1", "glass dish", ...)
`-- pick_place/
    |-- scene_spec.json             only with --ai-scene-agent
    |-- glb/
    |   |-- scene.usda  scene.pre_penetration_fix.usda
    |   `-- <label>/                asset.usd, manifest.json, mesh.glb, textures/
    |-- pick_place_scene.usda       scene as left by the demo
    |-- pick_place.mp4              final deliverable
    |-- penetration.json
    `-- pick_place.log  compose_isaac_scene.log  convert_asset.log  penetration.log
```

Key files

File	Content
`labels.json`	Label to sanitised-name mapping for all segmented classes.
`hull_consistency_rejections.json`	Per-class masks dropped by the 3D hull-consistency check.
`penetration.json`	Detected penetrating pairs and the lifts/settling applied.
`friction_assignments.json`	Per-object friction material (full-scene only).
`scene.usda` / `scene.pre_penetration_fix.usda`	Composed Isaac scene and its backup taken before the penetration fix.

4. CLI arguments

```
uv run python run_full_pipeline.py <image_folder> <scene_name> [options]
uv run python run_full_pipeline.py <video_file>   <scene_name> --video [options]
uv run python run_full_pipeline.py <scene_images> <scene_name> --mimic-robot episode.mp4

# pick-and-place with explicit targets
uv run python run_full_pipeline.py /path/to/images food2_vggt --classes "banana,bowl" \
    --mode pick-and-place --pick_place_target banana --place-target bowl
# full scene with robot
uv run python run_full_pipeline.py /path/to/images kitchen_floor --mode full-scene-with-robot \
    --classes "banana,bowl,plastic cup,white chair"
```

4.1 Positional arguments and input

Flag	Default	Description
`image_folder`	required	Folder of images, or a video file with **--video**.
`scene_name`	required	Name of the run; output goes to **runs/<scene_name>/**.
`--video`	off	Treat the first argument as a video: 5 fps, 960x540 frames in a temp folder. Not with **--stage-0-use-stereo**.
`--num_imgs_per_scene`	32	Number of views used by Stage 3 reconstruction.
`--skip-frames`	-1	Frame skipping passed to VGGT-Omega (-1 = none).
`--max-frames`	-1	Use only the first N images (sorted by name); applied before **--skip-frames**.
`--rotate-horizontal-deg`	0.0	Extra yaw applied after gravity alignment in Stage 0.
`--vggt-conf-thres`	20.0	Confidence threshold for Stage 0's exported point cloud.
`--vggt-viewer`	off	Spawn the rerun point-cloud viewer after Stage 0.
`--stage-0-use-stereo`	off	FoundationStereo instead of VGGT-Omega; needs exactly one left and one right image. Pick-and-place only; not with **--skip-frames**, **--max-frames**, **--vggt-viewer**.
`--stereo-conf`	*.conf in folder	ZED calibration file; requires **--stage-0-use-stereo**.

4.2 Mode and classes

Flag	Default	Description
`--mode`	pick-and-place	**pick-and-place**, **full-scene** or **full-scene-with-robot**.
`--classes`	none	Comma-separated labels ("banana,bowl"). Required for both full-scene modes; in pick-and-place it conditions Stage 0b discovery.
`--detection-box-padding-frac`	0.05	Pick-and-place: outward padding of each Stage 0b box before prompting SAM2.
`--depth_conf_thres`	50.0	Stage 1: depth confidence percentile filter.
`--depth_edge_rtol`	0.03	Stage 1: relative tolerance for removing depth-edge points.

4.3 MV-SAM3D (Stage 2)

Flag	Default	Description
`--mvsam3d-stage1-steps`	25	Shape inference steps (upstream default 50).
`--mvsam3d-stage2-steps`	12	Texture inference steps (upstream default 25).
`--mvsam3d-top-k-views`	5	Keep the k views that best cover each object; 0 or ≥ view count disables pruning.
`--mvsam3d-seed`	42	Diffusion random seed.
`--[no-]mvsam3d-align-shape-latents`	on	Align each view's canonical frame before fusing to avoid crossed slabs.

4.4 Reconstruction (Stages 3–12, full-scene)

Flag	Default	Description
`--fix_num_chunks`	16	Number of chunks for scene reconstruction.
`--max_chunks_per_group`	none	Limit chunks per group.
`--max_inflated_voxels`	none	Limit inflated voxels.
`--friction-table-path`	configs/materials/friction_table.example.yaml	Material friction table used in Stage 6.
`--ollama-model`	llama3.1:8b	Ollama model for friction inference.
`--robot-target`	random class	Class hit by the robot in Stage 12. Only with **full-scene-with-robot**; not with **--pick_place_target**, **--ai-scene-agent**.

4.5 Stage control and Isaac Sim

Flag	Default	Description
`--start-from-stage`	0	Skip stages below N, reusing outputs on disk.
`--stop-after-stage`	999	Exit after stage N.
`--skip_isaac`	off	Skip Stages 7–9b (USD conversion/composition). Not with **full-scene-with-robot**.
`--[no-]penetration-check`	on	Detect inter-object penetration after composing (about 10 s).
`--[no-]penetration-fix`	on	Write lifts that clear their pair back into scene.usda.
`--[no-]penetration-settle`	on	Settle dynamic objects under gravity after lifting; off = lift-only (capped at 20 mm).
`--room`	off	Wrap the table in a static kitchen room backdrop.
`--room-asset-dir`	assets/room	Override the room asset directory.

4.6 Pick-and-place targets

Flag	Default	Description
`--pick_place_target`	random	Object to pick; must exactly match a **--classes** label. Pick-and-place only. Without it (and without **--ai-scene-agent**) a random pair is chosen, which needs at least 2 classes.
`--place-target`	random	Object to place onto; needs **--pick_place_target**, differs from it, and is in **--classes**.
`--place-offset`	0.3,0.0,0.0	DX,DY,DZ (m) placement when there is no place target.
`--place-target-clearance`	0.05	Clearance above the place target (m).
`--gripper-open-width`	0.06	Gripper opening (m).
`--approach-side`	neg-y	**neg-x**, **pos-x**, **neg-y** or **pos-y**.
`--ai-scene-agent`	off	Stage P0 agent chooses targets and scene settings. Not with **--pick_place_target**, **--robot-target**.
`--scene-agent-ollama-model`	qwen2.5:7b	Ollama model for the scene agent.
`--mimic-robot VIDEO`	none	Episode video; its Qwen3-VL caption selects pick/place. Not with **--pick_place_target** or **--ai-scene-agent**.
`--caption-num-frames`	8	Frames sampled from the video for captioning (≥ 1).

Validation rules

- Full-scene modes require **--classes** and reject **--pick_place_target**, **--place-target** and **--ai-scene-agent**.
- **--robot-target** is only valid with **--mode full-scene-with-robot**; **full-scene-with-robot** cannot be combined with **--skip_isaac**.
- **--mimic-robot** and **--stage-0-use-stereo** are pick-and-place only; **--stereo-conf** requires **--stage-0-use-stereo**.